

Lakshya JEE 2.0 (2024)

Definite Integration

Assignment

Evaluate the following definite integrals.

1. $\int_0^1 \frac{\sin^{-1} \sqrt{x}}{\sqrt{x(1-x)}} dx$

2. $\int_0^{\ln 2} x e^{-x} dx$

3. $\int_0^e \left(\frac{1}{\sqrt{x \ln x}} + \sqrt{\frac{\ln x}{x}} \right) dx$

4. Given $f'(x) = \frac{\cos x}{x}$, $f\left(\frac{\pi}{2}\right) = a$, $f\left(\frac{3\pi}{2}\right) = b$. Find the value of the definite integral $\int_{\pi/2}^{3\pi/2} f(x) dx$.

5. $\int_{-1}^1 \frac{x dx}{\sqrt{5-4x}}$

6. $\int_2^e \left(\frac{1}{\ln x} - \frac{1}{\ln^2 x} \right) dx$

7. $\int_0^{\pi/4} \frac{\sin 2x}{\sin^4 x + \cos^4 x} dx$

8. $\int_2^{\pi/2} \frac{\cos x dx}{(1 + \sin x)(2 + \sin x)}$

9. $\int_0^{\pi/4} \frac{\sin^2 x \cdot \cos^2 x}{(\sin^3 x + \cos^3 x)^2} dx$

10. $\int_{1/3}^3 \frac{\sin^{-1} \frac{x}{\sqrt{1+x^2}}}{x} dx$

11. $\int_2^3 \frac{dx}{\sqrt{(x-1)(5-x)}}$

12. $\int_{3/2}^2 \left(\frac{x-1}{3-x} \right)^{1/2} dx$

13. $\int_0^{\pi/4} x \cos x \cos 3x dx$

14. $\int_0^{\pi/2} \frac{dx}{5 + 4 \sin x}$

15. $\int_2^3 \frac{dx}{(x-1)\sqrt{x^2-2x}}$

16. $\int_0^{\pi/2} \frac{dx}{1 + \cos \theta \cdot \cos x}$ $\theta \in (0, \pi)$

17. $\int_0^{\ln 3} \frac{e^x + 1}{e^{2x} + 1} dx$

18. $\int_0^{\pi/4} \cos 2x \sqrt{1 - \sin 2x} dx$

19. $\int_0^3 \sqrt{\frac{x}{3-x}} dx$

20. $\int_0^{1/2} \frac{dx}{(1-2x^2)\sqrt{1-x^2}}$

21. $\int_1^2 \frac{dx}{x(x^4 + 1)}$

22. $\int_0^{\pi/2} \sin \phi \cos \phi \sqrt{(a^2 \sin^2 \phi + b^2 \cos^2 \phi)} d\phi$

$a \neq b$ ($a > 0, b > 0$)

23. (a) $\int_0^{3\pi/4} ((1+x)\sin x + (1-x)\cos x) dx$

(b) $\int_{\pi/2}^{\pi} x^{\sin x} (1 + x \cos x \cdot \ln x + \sin x) dx$

24. $\int_0^1 x(\tan^{-1} x)^2 dx$

25. Suppose that f , f' and f'' are continuous on $[0, \ln 2]$ and that $f(0) = 0$, $f'(0) = 3$, $f(\ln 2) = 6$, $f'(\ln 2) = 4$ and $\int_0^{\ln 2} e^{-2x} \cdot f(x) dx = 3$. Find the value of

$$\int_0^{\ln 2} e^{-2x} \cdot f''(x) dx.$$

26. $\int_0^1 \frac{dx}{x^2 + 2x \cos \alpha + 1}$ where $-\pi < \alpha < \pi$

27. $\int_a^b \frac{dx}{\sqrt{1+x^2}}$ where $a = \frac{e-e^{-1}}{2}$ & $b = \frac{e^2-e^{-2}}{2}$

28. $\int_{0^+}^1 \frac{x^x (x^{2x} + 1)(\ln x + 1)}{x^{4x} + 1} dx$

29. $\int_0^1 x^5 \sqrt{\frac{1+x^2}{1-x^2}} dx$

30. Suppose that the function f , g , f' and g' are continuous over $[0, 1]$, $g(x) \neq 0$ for $x \in [0, 1]$, $f(0) = 0$, $g(0) = \pi$, $f(1) = \frac{2009}{2}$ and $g(1) = 1$.

31. $\int_0^{\pi/4} \frac{\sin \theta + \cos \theta}{9 + 16 \sin 2\theta} d\theta$

32. $\int_0^{\pi} \theta \sin^2 \theta \cos \theta d\theta$

33. $\int_0^{\pi/2} \frac{1 + 2 \cos x}{(2 + \cos x)^2} dx$

34. $\int_0^{\pi/2} \frac{x + \sin x}{1 + \cos x} dx$

35. Let $A = \int_{3/4}^{4/3} \frac{2x^2 + x + 1}{x^3 + x^2 + x + 1} dx$ then find the value of e^A .

36. $\int_0^1 \frac{2-x^2}{(1+x)\sqrt{1-x^2}} dx$

37. $\int_{-1}^1 \left(\frac{d}{dx} \left(\frac{1}{1+e^{1/x}} \right) \right) dx$

38. $\int_1^e \frac{dx}{\ln(x^x e^x)}$

39. $\int_0^{\pi} \left[\cos^2 \left(\frac{3\pi}{8} - \frac{x}{4} \right) - \cos^2 \left(\frac{11\pi}{8} + \frac{x}{4} \right) \right] dx$

40. If $f(\pi) = 2$ and $\int_0^{\pi} (f(x) + f''(x)) \sin x dx = 5$, then find $f(0)$.

41. $\int_a^b \frac{|x|}{x} dx$

42. $\int_{\ln 2}^{\ln 3} f(x) dx$, where $f(x) = e^{-x} + 2e^{-2x} + 3e^{-3x} + \dots \infty$

43. $\int_0^{\pi/2} \sqrt{\frac{\sec x - \tan x}{\sec x + \tan x}} \frac{\cos ec x}{\sqrt{1 + 2 \cos ec x}} dx$

44. $\int_0^1 x f''(x) dx$, where $f(x) = \cos(\tan^{-1} x)$



45. (a) If $g(x)$ is the inverse of $f(x)$ and $f(x)$ has domain $x \in [1, 5]$, where $f(1) = 2$ and $f(5) = 10$ then find the value of $\int_1^5 f(x)dx + \int_2^{10} g(y)dy$.
- (b) Suppose f is continuous, $f(0) = 0$, $f(1) = 1$, $f'(x) > 0$ and $\int_0^1 f(x)dx = \frac{1}{3}$. Find the value of the definite integral $\int_0^1 f^{-1}(y)dy$.



Note: Kindly find the Video Solution of DPPs Questions in the DPPs Section.

Answer Key

1. $\frac{\pi^2}{4}$

2. $\frac{1}{2} \ln\left(\frac{e}{2}\right)$

3. $2\sqrt{e}$

4. $2 - \frac{\pi}{2}(a - 3b)$

5. $\frac{1}{6}$

6. $e - \frac{2}{\ln 2}$

7. $\frac{\pi}{4}$

8. $\ln \frac{4}{3}$

9. $\frac{1}{6}$

10. $\frac{\pi \ln 3}{2}$

11. $\frac{\pi}{6}$

12. $\frac{\sqrt{3}}{2} - 1 + \frac{\pi}{6}$

13. $\frac{\pi - 3}{16}$

14. $\frac{2}{3} \tan^{-1} \frac{1}{3}$

15. $\frac{\pi}{3}$

16. $\frac{\theta}{\sin \theta}$

17. $\frac{1}{2} \left(\frac{\pi}{6} + \ln 3 - \ln 2 \right)$

18. $\frac{1}{3}$

19. $\frac{3\pi}{2}$

20. $\frac{1}{2} \ln(2 + \sqrt{3})$

21. $\frac{1}{4} \ln \frac{32}{17}$

22. $\frac{1}{3} \frac{a^3 - b^3}{a^2 - b^2}$

23. (a) $2(\sqrt{2} + 1)$ (b) $\left(\pi - \frac{\pi^2}{4} \right)$

24. $\frac{\pi}{4} \left(\frac{\pi}{4} - 1 \right) + \frac{1}{2} \ln 2$

25. **13**

26. $\frac{a}{2 \sin \alpha}$ if $a \neq 0$; $\frac{1}{2}$ if $\alpha = 0$

27. **1**

28. **0**

29. $\frac{3\pi + 8}{24}$

30. **2009**

31. $\frac{1}{20} \ln 3$

32. $-\frac{4}{9}$

33. $\frac{1}{2}$

34. $\frac{\pi}{2}$

35. $\frac{16}{9}$

36. $\frac{\pi}{2}$

37. $\frac{2}{1 + e}$

38. **$\ln 2$**



39. $\sqrt{2}$

40. 3

41. $|b| - |a|$

42. $\frac{1}{2}$

43. $\frac{\pi}{3}$

44. $1 - \frac{3}{2\sqrt{2}}$

45. (a) 48

(b) $\frac{2}{3}$

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